

Human Review: The Metrizable $C(K)$ Case of $\gamma^* = 1$

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct affirmative answer to the parenthesized metrizable version of Problem 7.3 in the source paper, namely that

$$\gamma^*(\mathcal{C}(\mathcal{K})) = 1$$

for compact metrizable spaces $\mathcal{K}$. It does not claim to settle the full nonmetrizable case.

The main idea is to decompose $\mathcal{K}$ into its Cantor–Bendixson perfect kernel P and the scattered remainder $\mathcal{K} \setminus P$. On the perfect part, P is compact metrizable and perfect, so $\mathcal{C}(P)$ is separable and octahedral. The source paper’s theorem for separable octahedral spaces then gives $\gamma^*(\mathcal{C}(P)) = 1$. Thus, for every small $\eta > 0$, there is a subgroup

$$D_P \subseteq \mathcal{C}(P)$$

which is 2-separated and $(1 + \eta)$ -dense.

The proof then uses the Dugundji extension theorem to choose a norm-one extension operator

$$E : \mathcal{C}(P) \longrightarrow \mathcal{C}(\mathcal{K}), \quad (Ed)|_P = d.$$

The remaining subgroup is

$$H = \{h \in \mathcal{C}(\mathcal{K}) : h|_P = 0 \text{ and } h(\mathcal{K}) \subseteq 2\mathbb{Z}\},$$

and the proposed packing subgroup is

$$D = E[D_P] + H.$$

The separation argument is straightforward and correct. If the D_P -component is nonzero, restriction to P detects norm at least 2. If the D_P -component vanishes but the H -component is nonzero, then the function takes a nonzero even-integer value, hence has norm at least 2.

The density argument also appears correct. Given $f \in \mathcal{C}(\mathcal{K})$, first approximate $f|_P$ by some $d \in D_P$, and then subtract the extension Ed . The remaining function

$$g = f - Ed$$

is already small on P . Therefore the set where $|g|$ is still too large lies inside $\mathcal{K} \setminus P$. This remainder is locally compact, metrizable and scattered, hence zero-dimensional. Consequently, the relevant compact subset of $\mathcal{K} \setminus P$ has a compact clopen neighbourhood $U \subseteq \mathcal{K} \setminus P$.

On this compact zero-dimensional set U , the proof applies exactly the even-integer approximation used in Theorem 4.4 of the source paper: take a finite clopen partition on which g has small oscillation, and on each clopen piece choose a nearest even integer to a representative value. This gives a continuous $2\mathbb{Z}$ -valued approximation on U , which can be extended by zero outside U . Since U is clopen and disjoint from P , the extension belongs to H . This completes the $(1 + \varepsilon)$ -density of D .

Review Outcome

Archive this packet as an apparently correct full solution to the metrizable case of the $C(K)$ problem. The proof is a clean gluing argument: the perfect kernel is handled by the source paper's separable-octahedral result, while the scattered remainder is handled by the source paper's zero-dimensional even-integer construction. The only points that should be cited carefully in a polished version are the zero-dimensionality of the locally compact metrizable scattered remainder and the use of Theorem 4.4 for the compact clopen neighbourhood U .